Assignment (Simulations for Fluid and Climate Technology, CM2014) VT2025

Instructions:

1. 2 - 3 students can work together.
2. You should produce Maple code for Q2, and a "pen-and-paper" solution for the Q3 task. The solution to Q3 can be hand-written, but make it legible and explain the steps with words. Bring a laptop, and make sure your Maple code runs.
3. For "Pass with excellence" and bonus points, you are expected to solve task Q4 and present it on the blackboard.
4. You can download Maple from https://www.kth.se/student/it/software/download/maple

Presentations:

The presentations of your work will be carried out during February 2025. Make sure your code runs! Please book a time slot in Canvas calendar. Questions to: ojs@kth.se

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Q1. Maple animations and solving differential equations

This assignment is for you to get comfortable with Maple.

i) There was a very nice Maple tutorial available written by a retired professor Armin: http://ingforum.haninge.kth.se/armin/MAPLE/dirmaple.html
But KTH decided to take it down. So I uploaded to Canvas a file called ***Maple help html.zip*** which contains some HTML-files called L8C, L8D, L9 and which contain code samples and animation samples. They are enough to understand the lab.

ii) Play around and construct some animations if you want, as described in file L8C above.

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Q2. Pathlines and streamlines

Definition: A **pathline** is the trajectory (path, curve) traveled by an individual fluid particle over some time period.
A **streamline** is a curve that is instantaneously (for the fixed time point) tangent to the velocity vector of the flow.

Consider the velocity field

$$\boldsymbol{V} = (0.6+0.4x+0.2y,\ 1.6+2.4\cos(5t) -0.8x -0.4y).$$

Please consult the content provided in Q1 for commands and examples on how to solve the below problems.

i) Find and plot the pathline through the point $A = (x(0), y(0)) = \left(\frac{2a+3}{a+1}, \frac{2b+3}{b+1} \right)$, where the constants *a* and *b* are two last digits in your ID number. Do not do this "by-hand"! Use Maple to compute the solution for you! Look in particular at Example 4 in file L9 to see how to do this.

ii) Animate the pathline. Look in particular at Example 5 in file L8C to see how to do this.

iii) For the fix time point, $t = 0$, find and plot the streamline through the point *A*. This is similar to part i), but now the time variable *t* is fixed. The solution is a curve, and it of course will be parametrized by some parameter (say) *s*. Make sure you understand that *s* does **not** represent time!

iv) Animate the velocity field ***V*** above.

v) Display the animation of the pathline from (ii) and animation of the velocity field ***V*** from part iv) in the same graph. The command *display* will be useful. Here is a description of the command https://www.maplesoft.com/support/help/Maple/view.aspx?path=plots%2Fdisplay. Perhaps it is easier to look at my example code PathLine.mw on Canvas.

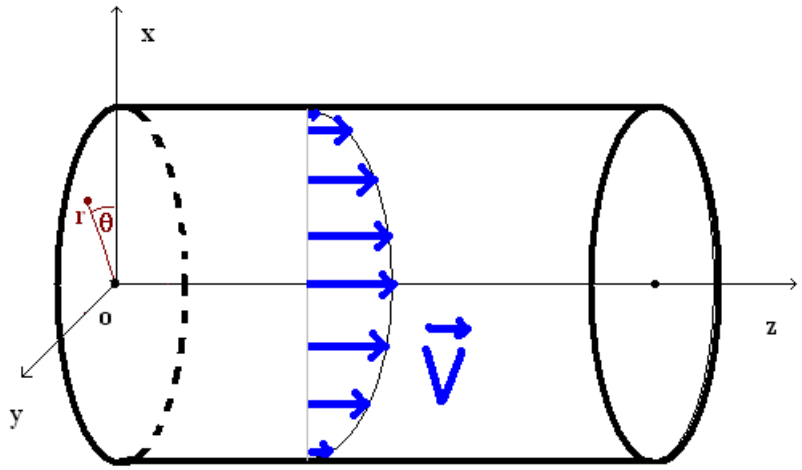
vi) If you have time and are bored, try to come up with some other time-dependent vector field ***V*** as above and animate the coolest pathline you can think of ☺.

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Q3. Computations with Navier-Stokes equations

Consider incompressible steady-state Newtonian flow in a long round pipe in the z-direction, with constant circular cross-section of radius R=4 m. We use the continuity and the Navier-Stokes equations (see the last page of this document for these equations) in cylindrical coordinates to find the velocity field $\mathbf{V}=(u_r, u_\theta, u_z)$ and the pressure field $P(r,\theta,z)$ if the fluid flow satisfies the following conditions:

c0. All partial derivatives with respect to time t are 0 (steady flow)
c1. μ=0.005 kg/(m·s) and ρ =1200 kg/m^3
c2. A Constant pressure gradient $\partial P/\partial z = -1/400$ Pa/m (caused by a pump) is applied in the horizontal axis (z-axis in our notation).
c3. The flow is parallel to the z axis, that is $u_r = 0$ and $u_\theta = 0$.
c4. We assume that the flow is axisymmetric. The velocity does not depend on θ, that is $\frac{\partial u_z}{\partial \theta} = 0$.
c5. Boundary condition 1 (No-slip boundary condition, $\mathbf{V}_{fluid}=\mathbf{V}_{wall}$): If r=4 then $u_z = 0$.
c6. Boundary condition 2: u_z has maximum at r=0 that is $\left.\frac{\partial u_z}{\partial r}\right|_{r=0} = 0$.



We choose x as a vertical axis, y and z are in a horizontal plane and the flow is parallel with the z-axis. We denote velocity vector **V**=(u_r, u_θ, u_z) where u_r, u_θ and u_z are r-component, θ-component and z-component in cylindrical coordinates. According to the assumptions we have u_r =0, u_θ = 0, and u_z does not depend on θ.

The tasks T1-T5 are to be solved using "pen and paper".

T1. Since x is the vertical axis we have that the vector **g**=(-g, 0,0) where g=9.81 m/s^2. Show that **g** in cylindrical coordinates has the following coordinates
$g_r = -g\cos\theta$, $g_\theta = g\sin\theta$ and $g_z = 0$

T2. Show that the continuity equation gives $\frac{\partial u_z}{\partial z} = 0$ that is show that u_z depends only on r.

T3. Denote $u_z = w(r)$ and simplify **eq b,c,d.**

T4. Find the pressure P.

T5. Find $u_z = w(r)$ and V = (0, 0, $w(r)$) [in cyl. coord].

The last task is to create an animation in Maple to visualize the solution. Nearly all the necessary code has been provided for you below. See "Help for T6".

T6. At t = 0 we consider the sphere $x^2+y^2+z^2 = 4$ (radius R1=2). Use Maple to show the 3D animation of the flow and shape changing of this sphere for $0 \le t \le 4$ *Explain (with few sentences) how this program works.*

T7. Change the radius of the sphere to R1 = 4 in your code, simulate, and explain what is happening.

Help for T6:
i) In the following program we used R=2, R1=1, w= r^2-4, tmax=2 and L=10 (length of the cylinder where we show the animation) .

ii) Change the values for R, R1, w, tmax and L (that is use the appropriate values from your task).

```
restart;
with(plots):
R:=2;R1:=1;tmax:=2; w:= 4-r^2; L:=10;

P1 := animate(plot3d, [[R1+sqrt(R1^2-
r^2)+t*w,r*cos(theta),r*sin(theta)], theta=0..2*Pi, r=0..R1
,numpoints=1000],t=0..tmax,frames=20,transparency=0.5):

P2 := animate(plot3d, [[R1-sqrt(R1^2-
r^2)+t*w,r*cos(theta),r*sin(theta)], theta=0..2*Pi,
r=0..R1,numpoints=1000],t=0..tmax,
frames=20,transparency=0.5):

Q := plot3d([u, R*cos(theta), R*sin(theta)],
theta=0..2*Pi,u=0..L,transparency=0.5):
display([P1,P2,
Q],style=patch,scaling=constrained,orientation=[-110,90]);
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Q4. Derivation of the Cauchy Momentum Equation

Armin (previous teacher) wrote a very nice derivation of the Cauchy Momentum Equation in the document found http://ingforum.haninge.kth.se/armin/AR_2000/HL2008/DERIV_NAVIER_STOKES.pdf (or on the Canvas page). For "pass with excellence", you should study this derivation and carry it out on the blackboard during your presentation. For a pass, you should showcase a deep understanding of the concepts and of the derivation.

The equations for Q3.

The continuity and the Navier-Stokes equations for an incompressible, isothermal Newtonian flow (that is, density ρ and viscosity μ are constants), with a velocity field $\vec{V} = (u_r, u_\theta, u_z)$ ***in cylindrical coordinates*** *(r, θ, z) :*

Incompressible continuity equation

$$\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{1}{r}\frac{\partial(u_\theta)}{\partial \theta} + \frac{\partial u_z}{\partial z} = 0 \qquad \textbf{eq a)}$$

Navier-Stokes equations in Cylindrical coordinates:

r-component:

$$\rho\left(\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta} - \frac{u_\theta^2}{r} + u_z\frac{\partial u_r}{\partial z}\right)$$
$$= -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_r}{\partial r}\right) - \frac{u_r}{r^2} + \frac{1}{r^2}\frac{\partial^2 u_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial^2 u_r}{\partial z^2}\right] \qquad \textbf{eq b)}$$

θ -component:

$$\rho\left(\frac{\partial u_\theta}{\partial t} + u_r\frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_r u_\theta}{r} + u_z\frac{\partial u_\theta}{\partial z}\right)$$
$$= -\frac{1}{r}\frac{\partial P}{\partial \theta} + \rho g_\theta + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_\theta}{\partial r}\right) - \frac{u_\theta}{r^2} + \frac{1}{r^2}\frac{\partial^2 u_\theta}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial u_r}{\partial \theta} + \frac{\partial^2 u_\theta}{\partial z^2}\right] \qquad \textbf{eq c)}$$

z-component:

$$\rho\left(\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta} + u_z\frac{\partial u_z}{\partial z}\right)$$
$$= -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2}\right] \qquad \textbf{eq d)}$$